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# Signal extraction and monitoring of motion loads based on wearable online device

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ABSTRACT

Monitoring of human exercise load is a hot area of wearable technology. The mainstream human motion calculation method predicts human motion through offline machine learning techniques. Personalized monitoring poses new challenges to the original learning model. Aiming at the detection of R peak value of motion load signal and the location of QRS complex, an improved R-peak detection algorithm based on adaptive threshold is proposed. The feature extraction method of motion load signal is introduced from multiple dimensions. The characteristics of time domain features and frequency domain features are analyzed. The time domain features and the eigenvalues of frequency domain features are defined and extracted. Experiments show that the proposed algorithm has higher detection accuracy. A multi-threshold-peak step algorithm is proposed and a motion state machine model is established. According to the periodic changes of the trunk movement and the arm swing acceleration in motion, the feature values are extracted, the step detection and the gait discrimination are performed, and the influence of external acceleration interference is excluded. The state machine adopts a nested structure, which is divided into two layers, a parent state and a child state, and calibrates the state transition condition. A verification method for the validity of feature parameters is proposed. The selected feature parameters are taken as an example to analyze and simulate the method. Simulation results show that the characteristic parameters in this paper are effective in the monitoring of the reasonableness of exercise load.

## 1. Introduction

In the field of sports, athletes must not only ensure a certain amount of exercise, but also ensure that the body is in the best condition. According to the traditional training method, it is subjectively judged whether the exercise meets the standard, and it is easy to cause the lack of exercise to achieve the effect of exercise or excessive exercise resulting in sports injury. The traditional intelligent device only uses the relationship between the number of steps of the human body and the parameters such as height and weight to estimate the energy consumption of the human body, and the calculation error is large [1–3]. The wearable exercise load monitoring system accurately calculates the exercise energy consumption by using the relationship between the acceleration and the exercise energy consumption during the movement of the human body, and can provide reliable quantitative data for the motion [4–6].

The step counter function is one of the essential functions of the motion load monitoring system [7]. The early pedometer was a mechanical pedometer. In fact, when the counterweight vibrates up and down, the balance is broken and a contact can be turned on and off [8]. The main function is completed by the electronic counter recording and display. Related scholars used a three-axis accelerometer to collect the acceleration signal of the wrist during walking, and designed a wrist pedometer, but the device has a single function and is not very applicable [9–11]. At present, there are two main types of counting algorithms in motion load monitoring. One is dynamic threshold method, which dynamically calculates the threshold within a certain range according to the acceleration data, and records it as one step when the acceleration data crosses the threshold [12]. This method uses the rising or falling region of the acceleration curve to make a judgment. The other is the peak detection method, which is to find the peak value of the signal based on the slope of the acceleration signal, calculate the number of motion steps by the number of effective peaks, and then judge the step from the inflection point position of the signal waveform [13]. It has also been mentioned that the positioning system of the smartphone is used to measure the distance of the user's movement, and then the number of steps is obtained by dividing the user's step size [14]. This is obviously cumbersome and has a large error. Today's wearable smart devices are moving in the direction of integrating multiple sensors, and it is foreseen that positioning technology will also be used in the field of motion step detection to improve the accuracy of detection [15]. The measurement of human exercise

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energy expenditure is the key to scientific movement [16]. If the energy consumption of the human body is not quantitatively and accurately monitored, the purpose of scientific sports cannot be achieved. In the field of sports and medicine, direct and indirect calorimetry are also used to detect human energy consumption. These two methods are mostly used as standards for human energy consumption, but the experimental conditions of calorimetry are complicated and inconvenient to operate [17]. None of these methods can meet the needs of athletes to detect sports energy in real time [18]. Accelerometer method is a method that uses the relationship between the acceleration signal integral and the oxygen consumption of human body motion to obtain the energy consumption of human body motion [19]. It is a relatively objective and simple detection method, and the accuracy of this method is better than the heart rate detection method. The footwork is high. Related scholars used a three-axis accelerometer to design a human body energy consumption detection system [20]. After acquiring human motion information, the acceleration feature quantity was used to realize the functions of human motion monitoring and exercise energy consumption calculation. Through the analysis of current research and product solutions, it is found that there are two main ways to achieve current motion monitoring. One is the detection method of dynamic threshold. The threshold is dynamically calculated based on the sampled data, and is recorded as an effective step when the signal data crosses the threshold. This method uses the rising or falling area of the waveform to make a judgment. The other method is a statistical peak detection method. This method uses the slope to find the peak value of the signal, and then refers to the signal waveform characteristics to find the effective signal peak, calculates the number of steps from the number of effective peaks, and then judges the step from the inflection point position of the signal waveform. However, the current product and algorithm research is mainly based on the signal processing of a single acceleration sensor, which has certain limitations in terms of scene discrimination and behavior discrimination.

At the device level, taking the friendly human–machine interaction as the starting point, this article selects wearable sensing devices that have the least interference with the user's lifestyle as a data acquisition tool, and uses a three-way acceleration sensor to collect signals. Specifically, the technical contributions of this article are summarized as follows:

First, the multi-domain feature extraction and analysis of exercise load signals are performed in this paper. For the extraction of the characteristics of the exercise load signal, it is necessary to select appropriate features for the monitoring of the exercise load signal according to the actual situation.

Second, based on the detection of QRS characteristic waveforms, this paper analyzes and extracts the time-domain and frequency-domain features of the exercise load signal. The motion monitoring algorithm is introduced. The algorithm is divided into two parts: step counting algorithm and motion monitoring state machine.

Third, an improved step monitoring algorithm is proposed, the design of the monitoring state machine is introduced, and the effectiveness of the characteristic parameters of multiple motion states is studied. A research method for the validity of acceleration signal characteristic parameters based on the monitoring algorithm is proposed, and the SVM (Support Vector Machine) monitoring algorithm is used to analyze and discuss the characteristic parameters involved.

The rest of this paper is organized as follows. Section 2 discusses the analysis and extraction of multi-domain features of the motion load signal. Section 3 examines the rationality monitoring of the amount of exercise load on wearable online devices. Finally, Section 4 summarizes the full text and points out the next direction of work.

## 2. Analysis and extraction of multi-domain characteristics of exercise load signal in wearable online equipment

### 2.1. Data collection for wearable online devices

Fig. 1 shows the system structure of data collection for wearable online devices. The exercise load signal data is first transmitted to the mobile device via the Bluetooth signal, and then transmitted and stored to the data collection device by wireless signals in accordance with a predetermined data format. Immediately after the signal of the exercise load is collected, the evaluation of the POMS scale is performed by the evaluation software installed in the mobile collection device, and then the emotion scale data of the subject is also uploaded to the data server. On the data server, the original data is filtered by wavelet analysis to filter out interference signals of power frequency interference and high frequency interference. Subsequently, the data characteristics of the motion load amount signal are extracted by a time–frequency analysis method. After that, the data characteristics of the exercise load signal and the emotion scale data are correlated, and the analysis results are compared and the analysis results are displayed in the form of a graph.

During the whole data acquisition process, the motion load signal is stored in the form of the original voltage value signal plus time information. The sampling frequency is 520 Hz. The motion load signal storage file is stored in the text file format with the subject number plus time information. The POMS gauge information is also stored in the text file format in the form of the subject number plus the evaluation time. The standardized storage process ensures the reliability and standardization of the data.

The data used in the algorithms involved in this paper are collected by sensing devices. The output signal of the sensor contains three components: x, y, and z in dynamic coordinates. The signals thus obtained are generated by the superposition of various human body movement effects and noise. The effects of body movements, arm movements, other subtle movements, and noise are reflected in the acquired signals. The walking frequency of a person is generally 110 steps/minute, and the slowest is 0.5 Hz. Even if the running frequency is less than 5 Hz, the frequency component of 0.5~5 Hz in the original signal can be regarded as a useful signal. According to the Nyquist sampling theorem, when the sampling frequency is greater than twice the highest frequency of the signal, the digital signal retains the information of the original signal intact. The sampling frequency of the signal is closely related to the preprocessing and the design of the algorithm. The larger the sampling frequency, the greater the order of the digital filter, and the larger the overall calculation amount. Therefore, in practical applications, the sampling frequency should be reduced as much as possible while ensuring that the target signal is not distorted. This design uses a sampling frequency of 20 Hz for data collection, which not only ensures that it can effectively reflect the movement state of the human body, but also reduces the computational complexity.

### 2.2. R-wave detection algorithm based on adaptive threshold

The R wave has the largest peak feature, which makes it significantly distinguishable from other waves. Therefore, the R wave is first detected before the QRS complex detection. In order to eliminate the influence of noise and to make the QRS waveform more prominent, this paper will first filter the ECG signal.

The amplitude of the R peak fluctuates within a certain range due to the influence of individuals and other external environments. Therefore, in order to avoid the missed detection caused by the amplitude fluctuation, in the process of performing the R wave detection, an adaptive threshold update mechanism needs to be introduced to realize the automatic adjustment of the threshold. We set the initial threshold $Th_{def}$ to 24% of the maximum amplitude over the 2.1s time period. When the magnitude of the motion load signal is greater than the initial

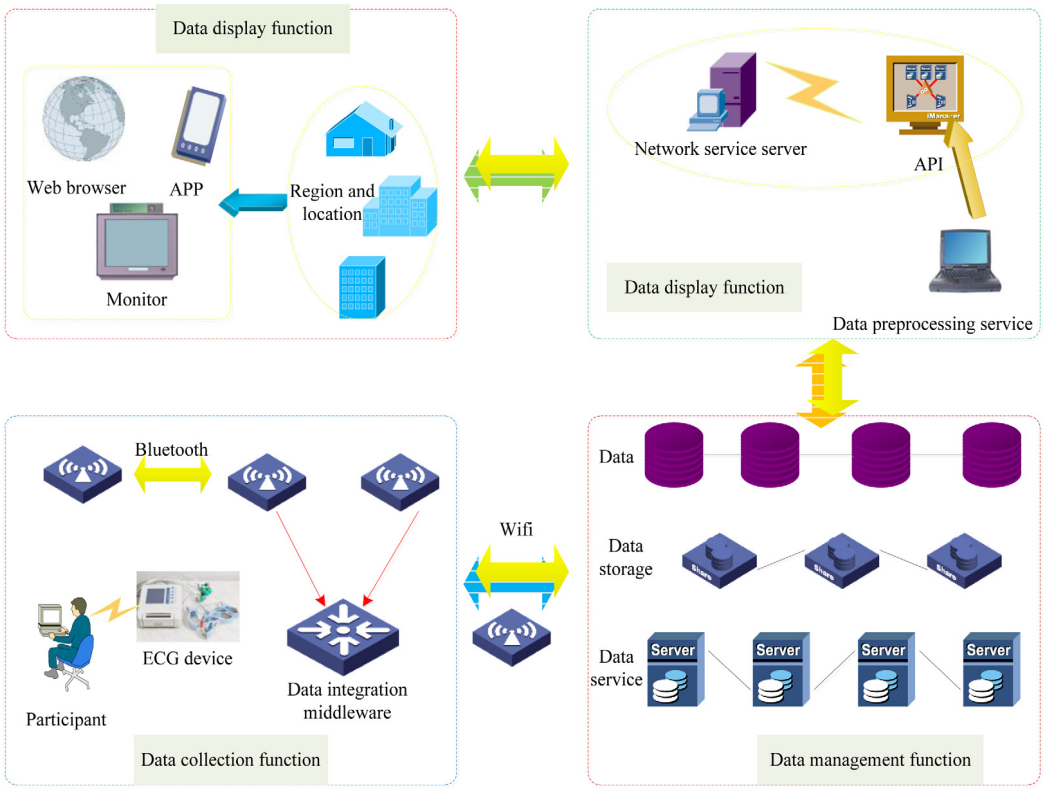



**Fig. 1.** System structure diagram of wearable online device data acquisition.

threshold $Th_{def}$, a new R wave peak is detected. The defined average threshold, $Th_{avg}$, is 24% of the most recently detected four peak mean values, as shown in the following equation:

$$Th_{avg} = 0.24 \times \frac{1}{4} \times \sum_{i<4} T_{k-i} \tag{1}$$

In order to avoid the situation that the R peak is not detected when the R peak amplitude is low due to the high threshold, an automatic threshold reduction mechanism is proposed. If no R-wave peak is detected within the average RR interval $RR_{avg}$ and the average threshold is greater than the set minimum threshold, the average threshold $Th_{avg}$ is reduced to 76% of the current value. The average RR interval $RR_{avg}$ is the mean of the four R peak intervals previously successfully detected, as shown in the following equation:

$$RR_{avg} = \frac{1}{4} \times \sum_{i=1}^{4} RR_i \tag{2}$$

The motion load signal after pre-processing is detected, and when the amplitude is higher than the average threshold $Th_{avg}$, the R peak detection algorithm is activated. The flow chart of the algorithm is shown in Fig. 2.

### 2.3. Simulation of the motion load signal R wave and QRS group positioning

In this paper, the motion load signal numbered 100 is tested, and the results of R wave detection are shown in Fig. 3. From Fig. 3, it can be clearly seen that all the R-wave peaks are detected and marked, and it can be intuitively proved that the adaptive threshold-based R-wave detection algorithm can accurately detect the R-wave peak.

In order to further verify the adaptive threshold-based R-wave detection algorithm proposed in this paper, a large number of data were experimentally verified. Among them, 30 of them were randomly selected from 50 samples, and the total R peak was 52,830. The test results are compared with the R wave results marked by the experts. In this paper, the classical evaluation criteria are used to calculate the experimental results, namely the sensitivity Se and the positive prediction rate + P (accuracy). The calculation formulas of the two are as follows:

**Table 1**
R Peak detection results statistics.

| Numbering | Actual R peak number | TP | FN | FP | Se | +P |
|---|---|---|---|---|---|---|
| 100 | 2271 | 2269 | 3 | 1 | 99 | 100 |
| 101 | 1859 | 1855 | 4 | 3 | 99 | 99 |
| 102 | 2188 | 2176 | 6 | 0 | 98 | 100 |
| 103 | 2079 | 2078 | 1 | 0 | 100 | 99 |
| 104 | 2231 | 2217 | 9 | 21 | 98 | 99 |
| 201 | 1961 | 1882 | 76 | 0 | 96 | 100 |
| 202 | 2011 | 1922 | 74 | 3 | 99 | 100 |
| 203 | 2978 | 2916 | 62 | 28 | 97 | 99 |
| 204 | 2123 | 2988 | 54 | 21 | 98 | 98 |
| 205 | 2633 | 2865 | 43 | 14 | 99 | 92 |
| 210 | 2433 | 2310 | 32 | 6 | 94 | 100 |
| 220 | 2155 | 2151 | 7 | 0 | 100 | 99 |
| Total | | | 169 | 53 | 99.02% | 99.87% |

$$Se = \frac{TP}{TP + FN} \tag{3}$$

$$+P = \frac{TP}{TP + FP} \tag{4}$$

Among them, TP indicates true positive, that is, the number of R peaks consistent with the actual labeling result, FN indicates false negative, that is, the number of R peak missed detection, and FP indicates false positive, that is, the number of false peaks of R peak. The experimental results are shown in Table 1.

It can be seen from the above table that the overall sensitivity of the R-peak detection algorithm based on adaptive threshold is 99.02%, and the positive prediction accuracy is 99.87%. The overall detection effect is better. Moreover, the adaptive threshold-based R-peak detection algorithm proposed in this paper is simple, and it is not suitable for

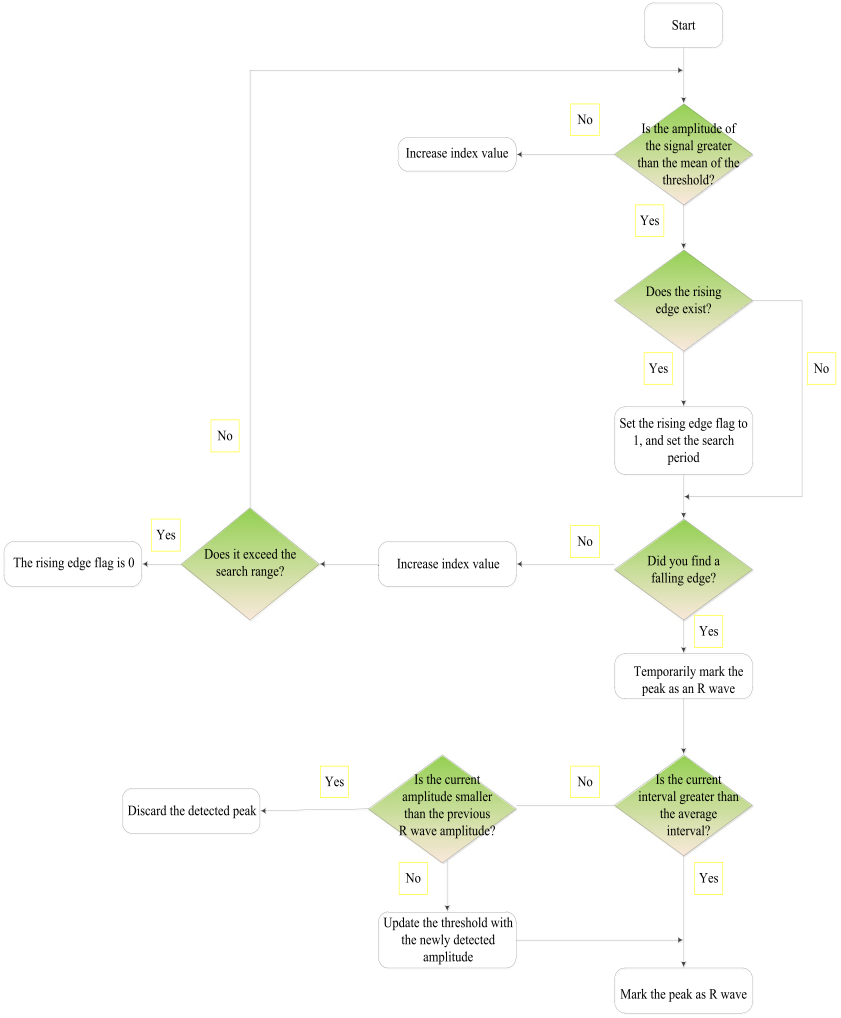



**Fig. 2.** R peak detection algorithm flow chart.

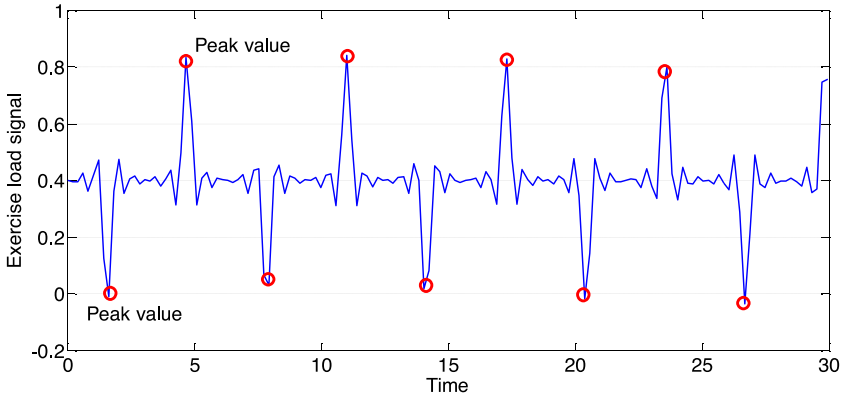



**Fig. 3.** R wave peak detection.

different individuals, and the threshold parameter is not required to be manually adjusted, and is more suitable for a wearable ECG monitoring device.

Based on the accurate detection of the R peak, the QRS complex is located according to the positional relationship of the Q wave and the S wave with respect to the R peak according to the R wave position as described in the paper. The experimental results are shown in Fig. 4. It can be seen that the algorithm described in this paper can accurately mark the position of the QRS complex.

### *2.4. Time-domain characteristics of the motion load signal*

The RR interval characteristics and QRS complex characteristics are the most intuitive and effective features for characterizing the time domain information of ECG signals. This paper defines and extracts the following RR interval features and QRS complex characteristics:

RR interval characteristics:

(1) Forward RR interval: The RR interval between the current R wave and its previous R wave is recorded as $RR_{pre}$;

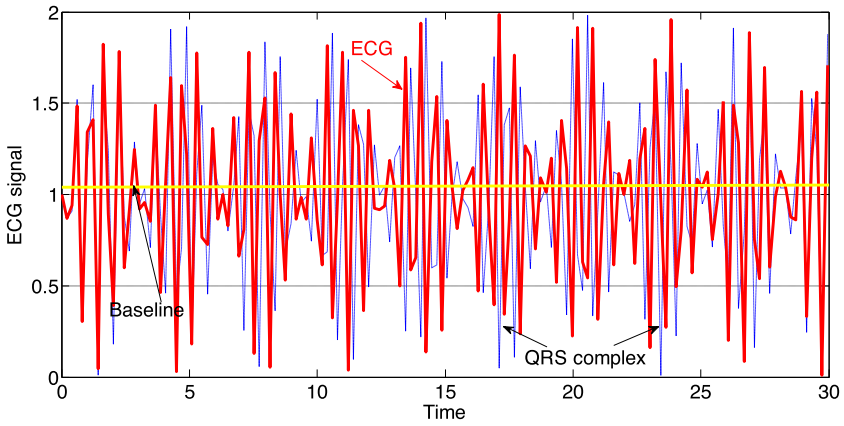



**Fig. 4.** Location of the QRS complex.

(2) Sub-term RR interval: The RR interval between the current R wave and the latter R wave is recorded as the $RR_{pos}$ interval;

(3) Average RR interval: The average of the last 10 RR intervals is recorded as $RR_{ave}$.

QRS complex characteristics:

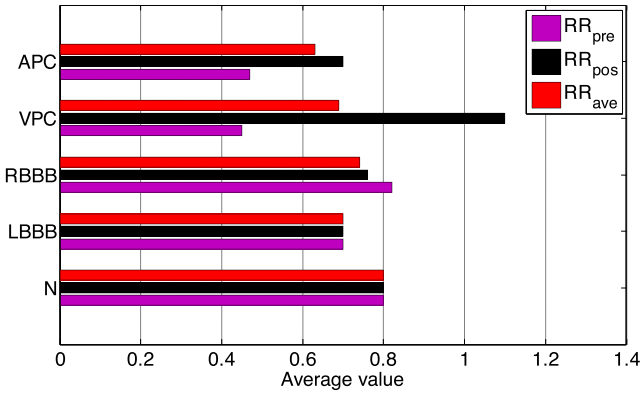
(1) R wave height: The height from the R peak to the zero potential reference point is denoted as $h_R$;

(2) QRS group height: The difference between the R peak height value and the Q wave height value is denoted as $QRS_h$.

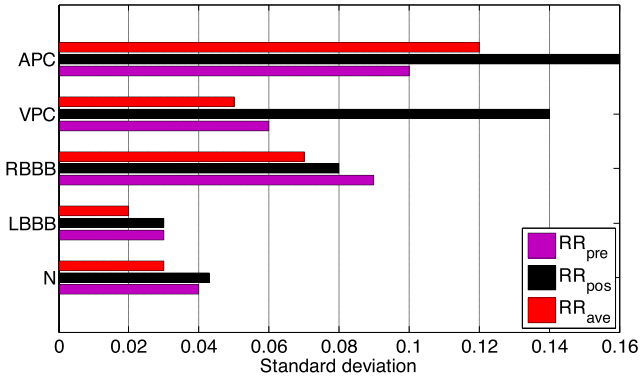
We use N for normal pulsation, LBBB for left bundle branch block, RBBB for right bundle branch block, VPC for ventricular premature beats, and APC for atrial premature beats. The time domain feature parameters defined above were extracted, and for each type of waveform, 600 cases were randomly selected and analyzed by taking the RR interval feature as an example.

It can be seen from Fig. 5 that the distribution of the N signal, LBBB signal and RBB signal is the most uniform, and the standard deviation of the RR interval of the VPC signal and the APC signal is much larger than the other two types of RR intervals. Overall, the same type of signal, RR interval features have good aggregation, and for different types of signals, RR interval features have a large difference, so the time domain features extracted in this paper can better reflect characteristics of the motion load signal.

A scatter plot of the N signal is plotted with three RR interval features as three-dimensional coordinates, as shown in Fig. 6. According to Fig. 6, it can be further seen that the time domain feature proposed in the present invention can reflect the characteristics of good N signal aggregation.




(a) Average of RR interval characteristics




(b) Standard deviation of RR interval characteristics

**Fig. 5.** Histogram of the five-phase heart beat RR interval feature.

### 2.5. Frequency domain characteristics of the motion load signal

Since the frequency domain information of the signal cannot be characterized by the time domain characteristics of the signal, and the frequency domain information contains more features that the time domain information cannot reflect, the frequency domain analysis method has been widely used in the field of bioelectric signals. This section will analyze the relevant features of the frequency domain of the motion load signal.

Let x(n) be a finite-length sequence of length N, then discrete Fourier transformation can be used to:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-\frac{2\pi}{N} nk} \tag{5}$$

Then the power spectral density of the signal x(n) can be expressed as:

$$P = \left| \frac{1}{\sqrt{2\pi}} \sum_{n=-\infty}^{\infty} x_n e^{-iwn} \right|^2 \tag{6}$$

where X(w) is the discrete Fourier transformation of x(n).

Compared to the traditional Fourier transform, the Fourier transform based on all-phase preprocessing does not suffer from severe waveform distortion and side line leakage. Therefore, this paper uses Fourier spectrum analysis based on all-phase preprocessing to perform power spectrum analysis on the motion load signal, which can more accurately reflect the information contained in the signal.

For different types of motion load signals, the power is concentrated in different frequency ranges. Therefore, the power spectral density of the motion load signal can be calculated, and the corresponding feature points can be extracted within a certain frequency range to form a feature vector.

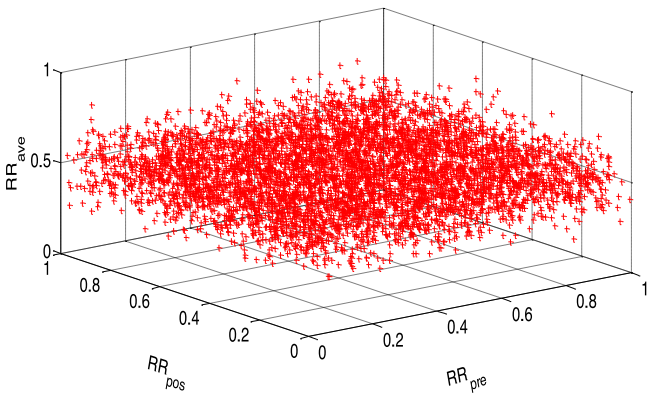



**Fig. 6.** N signal time domain feature scatter plot.

## 3. Reasonable monitoring of exercise load

### *3.1. Improved multi-threshold-peak algorithm and step-by-step monitoring simulation*

The difficulty of the time-counting algorithm for time-domain signal analysis is mainly the digital filtering in signal pre-processing and the discrimination of different motion states. The digital filtering method used in this paper can obtain waveforms with obvious and smooth features. The discrimination of different motion states can be understood as a problem of monitoring. According to the collected data, this paper divides the pace judgment into two categories: swing arm movement and non-swing arm movement. By comparing and analyzing the signals acquired by different motion states and other states, a multi-threshold-peak algorithm is designed. Multi-threshold is used to distinguish between different step states and other motion states, and effective step counting is performed by peak detection.

After preprocessing to obtain the sensor signal data sequence F(n), the target maximum value, the two minimum values before and after it, and the maximum value adjacent thereto are extracted from the data sequence. The difference between the ordinal numbers represents the interval time. This paper uses a sampling frequency of 25 Hz, and the interval between two adjacent data points is 40 ms.

After acquiring the feature value, the condition judgment phase is entered, and the threshold judgment and the cycle judgment are performed. The algorithm sets three-level step thresholds $T_1$, $T_2$, and $T_3$. Among them, $T_1$ is used as the entry condition for step determination, $T_2$ and $T_3$ are used as gait discrimination conditions, and two levels of abnormal thresholds $TR_1$ and $TR_2$ are set for motion abnormality monitoring. $TR_1$ is the lower threshold, $TR_2$ is the upper threshold, and $TR_2 < T_1 < TR_1$.

When an abnormal state occurs, the change of the generated acceleration is much larger than the change of the acceleration in the moving state. When the acceleration changes beyond the threshold $TR_1$, the abnormality determining program is entered, and when the peak value is reached after the peak value is reached, the monitoring state is entered. During the time, if the processing value of the acceleration does not exceed the threshold $TR_2$, it is regarded as a serious abnormal state, and the abnormal alarm program is triggered. If the value of the collected acceleration frequently exceeds the threshold value $TR_2$ during the monitoring time, it indicates that no damage or damage is caused, and the recovery motion can be regarded as a general abnormal state, and no alarm processing is performed. The anomaly discrimination process can be further determined by adjusting the threshold and setting a more detailed monitoring process.

If the difference between the peak value of the signal and the previous valley is greater than the step threshold $T_1$ and less than the abnormal threshold $TR_1$, the threshold condition of the step is satisfied.

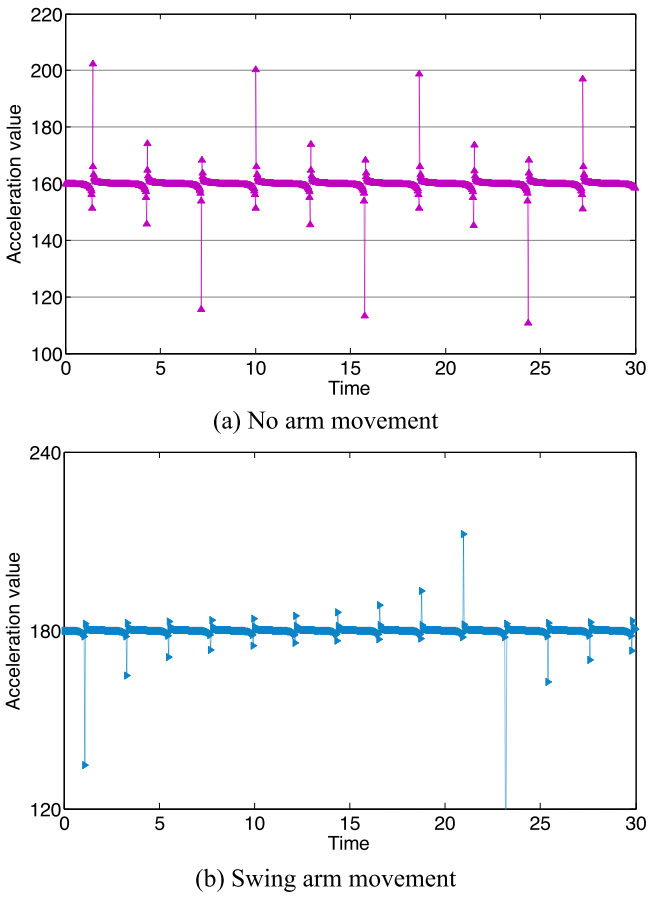



**Fig. 7.** Signal waveforms of two motion states.

We set the two-step gait thresholds $T_2$ and $T_3$ to distinguish the motion state of the arm swing.

As shown in Fig. 7, in the motion state of the arm swing, the acceleration change of the arm swing is not synchronized with the acceleration change of the trunk motion, resulting in a large difference in the acceleration signal of every two steps, and the difference is the same as the previous one. The peak time interval is shorter. This also makes the difference between the peaks and valleys much smaller than the difference between the peak and the previous valley, which provides conditions for gait discrimination. For a piece of step data, if the peak forward valley value and the backward peak-to-valley value are different, it means that the period of time is the motion state of the arm swing. If the peak forward valley value is not significantly different from the backward peak valley value, it means that this period of time is the motion state in which the arm does not swing. This judgment condition can also be replaced by the difference between adjacent valleys.

If there is a situation in which the trunk does not move and the arm swings (in situ swing arm), since the arm swing period is similar to the human body step period, the effective step condition can also be satisfied, as shown in Fig. 8. After analysis, it is found that, unlike the swing arm movement, the waveform exhibits a good cyclical trend during the stable in-situ swing arm process.

By observing and comparing the signal data in various scenes, it can be found that there are some isolated signal segments in the signals collected by irregular limb motion, which can satisfy both the threshold condition and the periodic threshold condition, but these signal fragments are mostly isolated. It does not have the periodic characteristics exhibited by the pace signal. Using this feature, set the number of consecutive steps as the starting condition of the counting

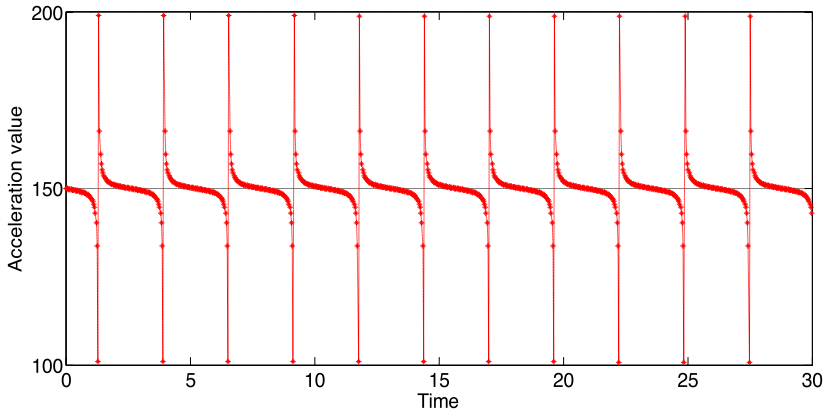



**Fig. 8.** In-situ swing arm signal waveform.

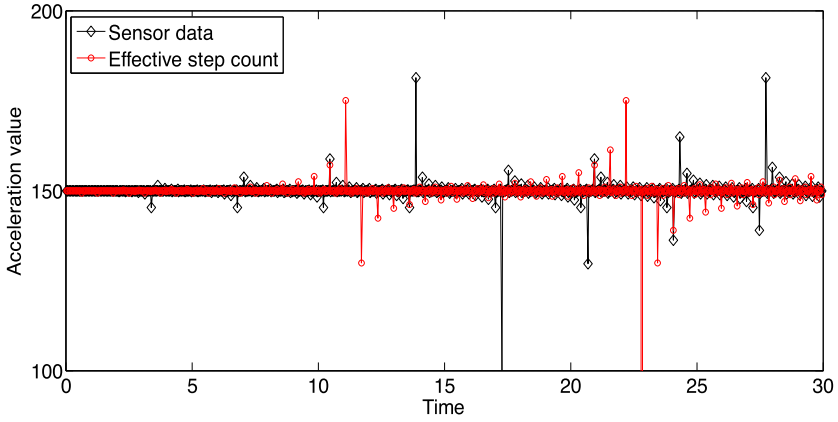



**Fig. 9.** Step detection results of the swing arm walking 50 steps.

algorithm. This article is set to three times. When the three consecutive steps are judged to be effective steps, it is determined that the signal acquisition time of the first effective step is the start time, enters the step state, and continues the normal step. If an invalid step occurs in the continuous step judgment of the initial stage, the effective step in the initial stage is discarded, and the continuous step judgment is repeated. In this way, the effects of such interference similar to the pace signal are eliminated.

The step detection time of the swing arm walking 50 steps is 30 s. From the simulation results in Fig. 9, in the case of the swing arm, the frequency component of the motion signal waveform is relatively simple and easy to detect. The test exercise time shown in Fig. 9 is 30 s. As shown in the coincident portion of the thick line with the ordinate of 150, most of the steps can be correctly monitored. In the results of a large number of simulation experiments, the pace of leakage that occurs in this gait can be almost ignored.

### 3.2. Monitoring state machine model

In order to clarify the relationship between various motion states and state transition conditions, this paper uses a nested state machine model to describe each state and its state transition, as shown in Fig. 10. Due to the large number of subdivisions of the system, it is difficult to avoid the disorder of the logical relationship by using the same state machine, so the monitoring method is adopted. Structurally, the motion state is divided into three parent states: static, motion and anomaly. And the states with similar logical relations or similar transition conditions are classified into the parent state, and each parent state can be regarded as a substate of the system alone. The machine organizes the state of the monitoring system in a nested state machine to form a clearly described motion monitoring state machine model.

**Table 2**
Parent state transition conditions.

| State transition | Transition condition |
|---|---|
| Static–Movement | Meet the number of consecutive steps |
| Movement–still | Determine the invalid pace |
| Stationary–abnormal | Satisfying the abnormal threshold condition |
| Abnormal–stationary | Jump out of exception handler |
| Movement–abnormal | Satisfying the abnormal threshold condition |

The lower 8 bits and the upper 8 bits of the identifier are used to identify the parent and child states, respectively. The state transition conditions of the parent state are shown in Table 2.

The stationary mother state is the initial state after the system is started. After entering, it will enter the transient static substate by default. If the time in this state exceeds a threshold, it will transit to the long-term stationary substate and perform corresponding prompt processing.

In the state of the sports mother, there are three sub-states of no arm swing, swing arm movement and in-situ swing arm. Due to the existence of a certain amount of exercise in the original swing arm, it is temporarily classified as such. Each effective step judgment must be classified into one of the three.

Once the abnormal threshold $TR_1$ condition is met, the possible abnormal state is entered, and then the abnormality degree is judged by the abnormality discriminating program, and corresponding processing is performed. After the exception handler is completed, the jump goes to a standstill.

The motion monitoring state machine essentially divides the sensor signal into a large number of time segments in the time domain, and

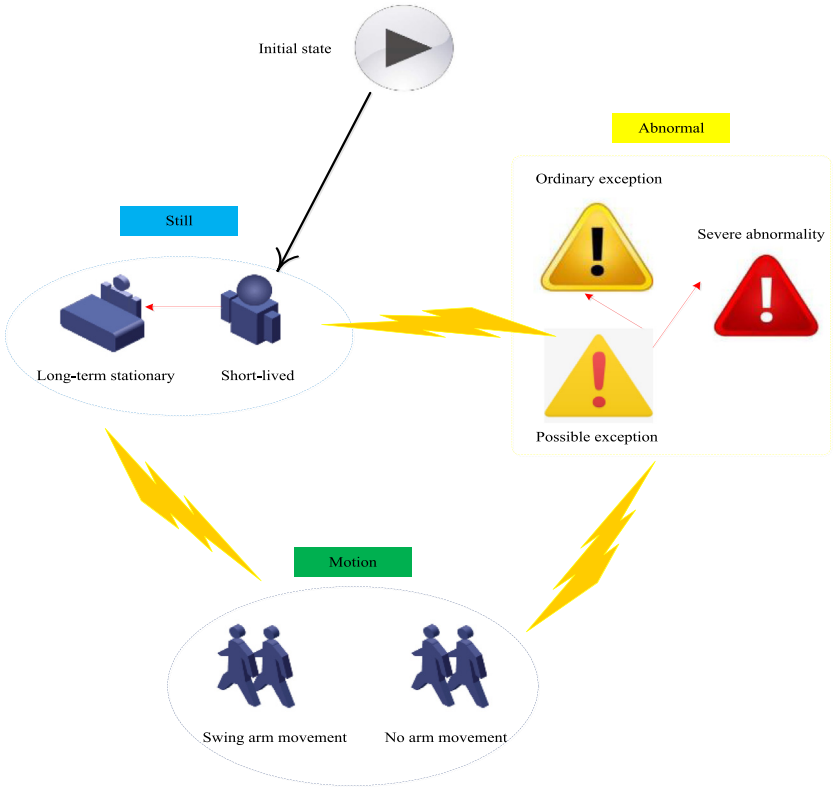



**Fig. 10.** Schematic of motion monitoring state machine.

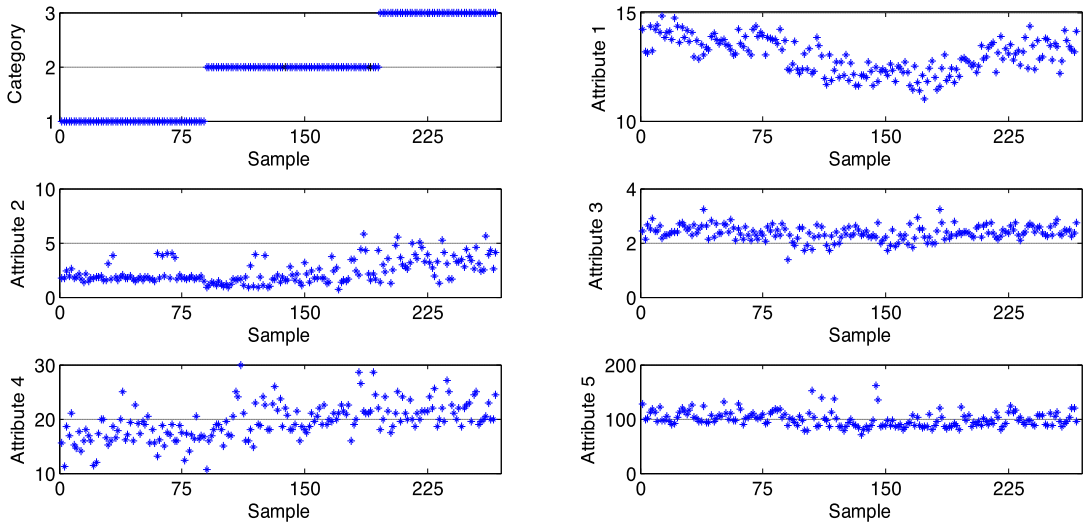


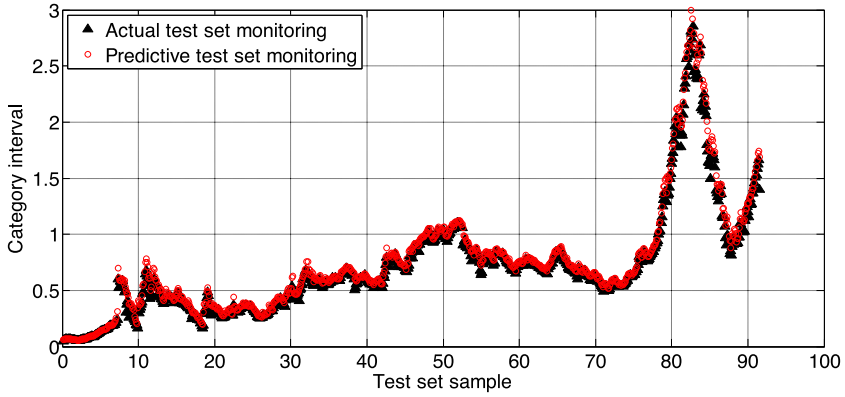

**Fig. 11.** Training set attribute fractal dimension visualization.

monitors the signal segments according to the characteristic parameters. The accuracy of the monitoring can reflect the validity of the selected characteristic parameters to a certain extent.

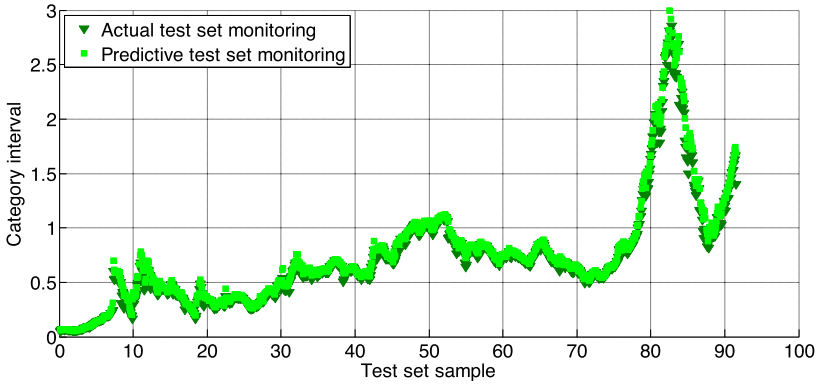
### 3.3. Feature parameter validation method

The step-counting algorithm in this paper can better distinguish the motion state from other states, but it cannot reflect the role of the selected feature parameters in the determination of the specific motion state (including the parent state and the sub-state), that is, the selected feature parameters cannot be verified. Aiming at this problem, a method to study the validity of feature parameters by means of monitoring algorithm is found. Although monitoring algorithms are difficult to directly apply to the monitoring of motion state, the selected feature parameters can be analyzed by means of a monitoring algorithm. The validity of the feature parameters determines the effectiveness of the monitoring algorithm in the specific judgment link to a certain extent.

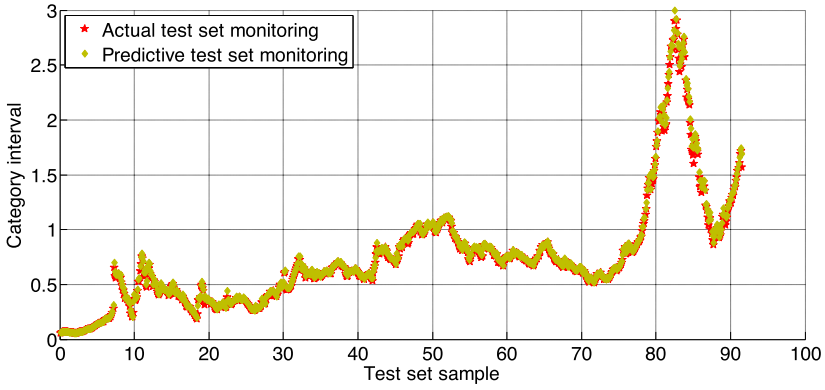
In addition to the abnormal state, the determination of the remaining states is performed through the same program branch, and the same or part of the same feature parameters are judged. It can be considered that these states are in the same domain, and can be used as the SVM category, and different labels are assigned. Among the characteristic parameters used in this paper, the number of consecutive steps is based on counting statistics, while the peak-to-valley value, the step-period, etc. can be regarded as time-domain parameters. The two dimensions are different, and the number of consecutive steps cannot




(a) Full parameter monitoring results




(b) Monitoring results of the masking period parameters




(c) Monitoring results of the peak-to-valley parameter

**Fig. 12.** Monitoring results of parameters.

directly describe the sampling points. Therefore, only the time domain parameters are selected as the training set attributes (dimensions).

This SVM model uses the swing arm motion, the non-swing arm motion, and the sit-down as the categories, corresponding to the category labels 1∼3.

The training set attributes include

(1) Peak and valley value (the difference between the target peak value and the forward valley value);

(2) Valley value (the difference between the two valley values before and after the target peak value);

(3) Peak-to-peak (the difference between the target peak and the backward adjacent peak);

(4) Step period (the number of data points between the two adjacent valley values before and after the target peak);

(5) Cycles of adjacent steps.

From the data collected multiple times, 260 samples are randomly selected for each category as the training set. The categories and attribute characteristics of the training set data are shown in Fig. 11. The training set is input into Libsvm's one-to-one multi-monitoring algorithm training monitoring model. A further 92 samples were selected as test sets for each category, and the test set was predicted using the monitoring model to obtain the predicted results, as shown in Fig. 12.

By changing the combination of characteristic parameters, the results of monitoring with single or multiple characteristic parameters can be obtained. Comparing the results of different parameter combinations, the effectiveness of each characteristic parameter can be evaluated from the monitoring accuracy. Of course, the results obtained by this method are directly related to the training samples. The larger the number of training samples and the wider the sample source, the higher the confidence of the method.

### 3.4. Simulation of the rationality monitoring of exercise load

This verification uses the C-SVC model. The selected kernel function is a radial basis kernel function. Therefore, the final decision function of each of the two monitors is:

$$y_i(x) = \operatorname{sgn}\left[\sum_{i=1}^{n} w_i \exp(-\gamma \left\|x_i - x\right\|^2) + b\right] \tag{7}$$

where $\|x_i - x\|$ is the two norm distance, n is the number of support vector machines, $w_i$ represents the coefficient of the support vector, and $x_i$ represents the support vector.

It can be seen from Fig. 12 that in three cases, the monitoring of the parameters has a good overall effect. Therefore, the motion monitoring algorithm of this paper has better monitoring effect of the pace, which can distinguish the swing arm movement, the swing arm movement and the sit-down. It can be seen from Fig. 12(a) and (b) that the monitoring result after the masking period parameter is compared with the case of the full parameter, the monitoring of the step state does not change, and the influence of the period parameter on the motion state monitoring in the algorithm can be smaller. It can be seen from Fig. 12(c) that after the peak-to-valley parameter is masked, some non-step test samples are erroneously monitored into the step category, and the peak-to-valley value is an important step determination condition. It can be seen that the monitoring effect of the step sample and the non-step sample is still good only by using the peak-to-valley value. Therefore, the peak-to-valley parameter is highly effective for monitoring the pace. Referring to all the results obtained in this paper, it can be concluded that among the selected characteristic parameters of the motion monitoring algorithm, the peak-to-valley parameters also have an important influence on the effectiveness of the step detection.

Under the condition of this system, the periodic parameters have little effect on the motion monitoring. The peak-to-valley parameter is the most important condition for the step detection. The valley parameter has a certain monitoring effect on the monitoring of the swing arm or not. Parameters have a major impact on gait monitoring. Of course, these judgments are based on the characteristics of the sample set, and as the size of the sample set expands, other conclusions may be obtained. The representativeness and quantity of the training samples have a great impact on the method.

## 4. Conclusion

This paper adopts an easy-to-implement method based on filtering and threshold detection to detect the R-wave of the exercise load signal, and improves the threshold adaptive adjustment method. The position of the QRS complex was determined by a position-based method. The experimental results show that the improved method proposed in this paper can accurately detect the R wave and locate the QRS complex. Time-domain features such as R-R interval and R-wave height are extracted, and the power spectral density of several types of exercise load is analyzed. From the power spectral density, we can see the difference in the frequency domain between different types of exercise load. Parameters such as peak-to-valley value, peak-to-peak value, step period, and number of consecutive steps were selected as characteristic parameters. An improved multi-threshold-peak algorithm was designed and verified by simulation. The algorithm can accurately monitor the reasonableness of the step load, and can eliminate the influence of external interference, reducing the probability of false monitoring. The design of the monitoring state machine is introduced, the problem of selecting the characteristic parameters for monitoring the motion state is analyzed, and the SVM multi-monitoring algorithm is used for simulation verification. However, the selection of the characteristic parameters and the setting of the threshold in the step counting algorithm require a large amount of test data to support, in order to further improve the accuracy of the step counting algorithm, more data can be tested and analyzed in the next step, thereby obtaining the best feature parameters and thresholds.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Xidan Gong:** Data curation. **Huichao He:** Formal analysis.

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